

Causal Modelling

Pearl's Ladder of Causation

- 1) Association $P(Y|X)$
- 2) Intervention $P(Y|\text{do}(X))$
- 3) Counterfactuals $P(Y|X, \text{do}(X'))$

variables are random variables

A structural causal model (SCM) consists of a graph on endogenous (observed) variables X_i , together with exogenous (unobserved) variables U_i that are jointly independent, and structural assignments $\leftarrow$ parents of X_i .

$$X_i = f_i(PA_i, U_i)$$

We will assume that the SCM is a directed acyclic graph.

Example: A = Aspirine, H = Headache, U is some genetic factor s.t.

$$U = \begin{cases} 1 & \text{if } A \text{ is effective} \\ 0 & \text{o.w.} \end{cases}$$

Then $H = (1-A) \vee (1-U)$.

We can calculate from observation $P(H=0|A=1)$. Note that this might be greater than $P(H=0)$ or $P(H=0|A=0)$ because of confounders.

We can also calculate $P(H=0|\text{do}(A=1))$ from e.g. a randomised control trial.

We get $H = (1-1) \vee (1-U) = 1-U$,

so now H depends on the population and confounding is broken.

Finally, we can calculate counterfactuals $P(H|A=1, H=0, \text{do}(A=0))$. We see that $U=1$, hence $H = (1-A) \vee (1-U) = 1$.

Removing the structural assignments, we get a causal graph. Causal graphs can calculate associations and interventions, but not counterfactuals. Generally, interventions can be mapped to pure observations via the do-calculus.

Let G be a DAG. For a set of nodes X let $G_{\bar{X}}$ denote the graph deleting all incoming edges and G_X outgoing edges. We call Y, Z d-separated given W , written $(Y \perp\!\!\!\perp Z | W)_G$, if every undirected path from Y to Z is blocked by W , i.e. if any path from Y to Z contains either

- $\rightarrow Z \rightarrow$, or
- $\leftarrow Z \rightarrow$, or
- $\rightarrow V \leftarrow$ with $V \notin Z$ and no successor of V is in Z .

Theorem (do-calculus): Let X, Y, Z, W be disjoint. Then

$$1) P(Y|\text{do}(X), Z, W) = P(Y|\text{do}(X), W) \text{ if } (Y \perp\!\!\!\perp Z | X, W)_{G_{\bar{X}}}$$

Insertion/deletion of observations
If Z is d-separated from Y when X is intervened on then observing Z gives no further information

$$2) P(Y|\text{do}(X), \text{do}(Z), W) = P(Y|\text{do}(X), Z, W) \text{ if } (Y \perp\!\!\!\perp Z | X, W)_{G_{\bar{X}, Z}}$$

Action/observation exchange
If all backdoor paths from Z to Y are blocked then interventions and observations on Z give the same observations

$$3) P(Y|\text{do}(X), \text{do}(Z), W) = P(Y|\text{do}(X), W) \text{ if } (Y \perp\!\!\!\perp Z | X, W)_{G_{\bar{X}, Z, W}}$$

Insertion/deletion of actions
Intervening on Z has no effect on Y if there is no causal path to Y or W

Furthermore, these three rules are complete, i.e. an intervention is identifiable from observational data iff the do-calculus reduces it to an expression without do's.

Example (Backdoor criterion): Let Z satisfy the backdoor criterion for (X, Y) , i.e.

- no node in Z descends from X , and
 - Z blocks all paths between X and Y containing an arrow into X .
- Then

$$\begin{aligned} P(Y|\text{do}(X)) &= \int_Z P(Y|\text{do}(X), z) P(z|\text{do}(X)) \\ &\stackrel{\text{Rule 3}}{=} \int_Z P(Y|\text{do}(X), z) P(z) \\ &\stackrel{\text{Rule 2}}{=} \int_Z P(Y|X, z) P(z) \end{aligned}$$